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Shaker for aerosol spray cans

Abstract

A device for shaking aerosol spray cans (**2**) and spray paint containers that agitate and thoroughly mixes the contents contained therein by replicating the up-and-down and back-and-forth spinning motions an individual normally performs with his or her hand to shake a spray can. The spray can shaker (**1**) of the present invention is adaptable to multiple sizes and shapes and spray cans.

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Background/Summary

FIELD OF THE INVENTION

(1) This invention relates to devices used to mix flowable liquids contained in pressurized aerosol canisters, and more particularly, a device for shaking a spray can to agitate and thoroughly mix the contents contained inside of the spray can

BACKGROUND OF THE INVENTION

(2) A There are a plurality of spray products on the market that require shaking to ensure any liquids and propellant contained in the canister are well mixed prior to being sprayed out of a

pushbutton or trigger activated nozzle. Many propellants used in aerosols aren't miscible in other liquids, thus, the liquids and propellants will form separate layers in the canister when stored. This may also happen with the ingredients of the liquid product itself. If not shaken prior to use, the canister will expel excess propellant, an uneven spray, and/or nothing at all.
(3) Spray paint is perhaps the most well-known aerosol product that requires vigorous shaking prior to use to ensure the contents are properly mixed with each other prior to application. Failure to thoroughly shake a paint will cause an uneven finish when applying canisters A the paint. Shaking paint cans and other aerosol canisters can be difficult for individuals who suffer from dexterity issues, such as carpel tunnel disease. This is especially true of paint cans that have been stored for any extended periods of time.
(4) Therefore, a need exists for a device for shaking spray paint cans and other spray cans to agitate and thoroughly mix the contents contained therein.
SUMMARY OF THE INVENTION
(5) The primary object of the present invention is to provide a device for shaking paint cans and other spray cans to agitate and thoroughly mix the contents contained therein prior to application.
(6) An additional object of the present invention is to provide a device for shaking paint cans and other aerosol containers that is adaptable to multiple sizes of spray cans.
(7) An additional object of the present invention is to provide a device for shaking paint cans and other spray cans that is adaptable to aerosol canisters having different shapes.
(8) The present invention fulfills the above and other objects by providing a spray can shaker having a carrier wherein a spray canister is secured in the shaker. The carrier is slidably attached to guides that are supported on opposing sides of the carrier by vertical supports and a stationary base having a non-slip bottom surface. A stationary handle may extend above the spray can shaker to provide a contact point for a user to hold the spray can shaker in a fixed position on a flat working surface.
(9) A manual or motorized power supply is used to move the carrier and secured spray can in an oscillating forward and backward motion. A manual power supply is illustrated herein having a crank handle connected to a gear box and a flywheel with a counterweight that causes the carrier to replicate an up-and-down and turning motion an individual normally performs with his or her hand to mix a spray can.
(10) The above and other objects, features and advantages of the present invention should become even more readily apparent to those skilled in the art upon a reading of the following detailed description in conjunction with the drawings wherein there is shown and described illustrative embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the following detailed description, reference will be made to the attached drawings in which:
(2) FIG. **1** is a top perspective view of an empty spray can shaker of the present invention;
(3) FIG. **2** is a top view of the spray can shaker of the present invention having a spray can secured therein;
(4) FIG. **3** is a rear view of the spray can shaker of the present invention;
(5) FIG. **4** is a sectional side view along lines A-A of FIG. **3** showing the flywheel and counterweight used to assist in actuating the carrier;
(6) FIG. **5** is a sectional side view along lines B-B of FIG. **3** providing an internal view of the gear box used to assist in actuating the carrier;
(7) FIG. **6** is a right side view of the spray can shaker of the present invention;
(8) FIG. **7** is a side view of a carrier of the present invention;

(9) FIG. **8** is a top view of a carrier of the present invention;
(10) FIG. **9** is a top view of a carrier of the present invention accommodating a short spray can; and
(11) FIG. **10** is an end view of a C-shaped clip from the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(12) For purposes of describing the preferred embodiment, the terminology used in reference to the numbered accessories in the drawings is as follows: **1**. spray can shaker, generally **2**. spray can **3**. base **4**. bottom surface of base **5**. top surface of base **6**. front edge of base **7**. back edge of base **8**. right side edge of base **9**. left side edge of base **10**. right vertical side support **11**. left vertical side support **12**. carrier **13**. right side guide **14**. left side guide **15**. pin **16**. power supply **17**. crank handle **18**. gear box **19**. gear **20**. fly wheel **21**. counterweight **22**. carrier body **23**. front end of body **24**. back end of body **25**. inner surface of body **26**. outer surface of body **27**. clip, generally **28**. top clip **29**. bottom clip **30**. middle clip **31**. step **32**. stop **33**. stationary handle

(13) With reference to FIG. **1** and FIG. **2**, views of a spray can shaker **1** of the present invention without a spray can **2** and with a spray can **2**, respectively, are illustrated. The spray can shaker **1** comprises a base **3** having a planar bottom surface **4**, top surface **5**, front A edge **6**, back edge **7**, a right side edge **8**, and a left side edge **9**. A non-slip surface and or/means to mechanically mount the base **3** to a work surface is preferably located on the bottom surface **4** of the base **3** to keep the base stationary during use.

(14) A right vertical side supports **10** and a left vertical support **11** each extend upward at an angle, respectively, from the right side edge **8** and a left side edge **9**.

(15) The right vertical side support **10** and the left vertical support **11** support a carrier **12** in an A elevated position above the base **3**. The carrier **12** provides means for securing a spray can **2** in the spray can shaker **1**. The carrier **12** is slidably attached to the right side vertical support **10** and the left vertical support **11** via a right side guide **13** and a left side guide **14**, both of which preferably comprise fixed channels that engage pins **15** extending from a right side of the carrier **12** and a left side of the carrier **12**.

(16) A motorized or a manual power supply **16** may be used to move the carrier **12** and a spray can **2** secured therein in the carrier **12** in a front-to-back oscillating motion in relation to the base **3**, thereby replicating an up-and-down and turning motion an individual normally performs with his or her hand to mix a spray can **2**. A stationary handle **33** preferably extends above the spray can shaker **1** in a substantially centered position to provide a contact point for a user to hold the spray can shaker **1** a fixed position on a work surface.

(17) With reference to FIGS. **3-6**, the manual power supply **16** is illustrated herein having a crank handle **17** connected to a gear box **18** having at least two gears **19** housed therein to transfer the power from the rotating crank handle **17** to a flywheel **20** having a counterweight **21** located thereon that is pivotally attached to a back end **24** of the carrier **3** in a circular motion combined with a front-to-back motion in relation to the base **3**, thereby replicating an up-and-down and turning motion an individual normally performs with his or her hand to mix a spray can **2**.

(18) With reference to FIGS. **7** and **8**, a side view and a top view, respectively, of a carrier **12** of the present invention is illustrated. The carrier **12** comprises a preferably open tubular-shaped body **22** having a front end **23**, a back end **24**, an inner surface **25**, and an outer surface **26**. At least one C-shaped clip **27** is located within the tubular-shaped body **22** and is used to grasp and lock a spray can **2** into carrier the spray can shaker **1**. As illustrated herein the carrier **12** comprises a top clip **28** located on the front end **23** of the body **22**, a bottom clip **29** located on the back end **24** of the body **22**, and at least one middle clip **30** located between the front end **23** of the body **22** and the back end **24** of the body **22**.

(19) Each clip **27** may act as an attachment means for securing a spray can **2** and may also act as ledges or stops so the carrier **12** may accommodate different sizes of spray cans **2**. For example, when a full-sized spray can **2** may be inserted into the carrier **12**, as illustrated in FIG. **2**, it engages all of the clips **27** located in the carrier **12**. When a short spray can **2** is inserted into the carrier **12**,

as illustrated in FIG. **9**, it is not long enough to engage the bottom clip **29** and only engages the top clip **28** and two middle clips **30** with one of the middle clips **30** acting support for the bottom of the spray can **2**.

(20) As illustrated in FIG. **9**, each clip **27** may comprise an indented step **31**. Each indented step **31** may act as a stop **32** to engage top and bottom perimeter edges of a spray can **2**. For example, as illustrated in FIG. **10**, a step **31** located on a middle clip is engaging a bottom perimeter edge of the spray can **2** and acting as a stop **32** that keeps a top perimeter edge of the spray can **2** locked in secure contact with a step **31** located on the top clip **28**.

(21) It is to be understood that while a preferred embodiment of the invention is illustrated, it is not to be limited to the specific form or arrangement of parts herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention and the invention is not to be considered limited to what is shown and described in the specification and drawings.

Claims

1. A spray can shaker comprising: a base having a bottom surface, top surface, front edge, back edge, a right side edge, and a left side edge; a right vertical side support extending upward at an angle from the right side edge of the base; a left vertical side support extending upward at an angle from the left side edge of the base; said right vertical side support and left vertical side support supporting a carrier in an elevated position above the base; said carrier providing a means for securing a spray can in the carrier; said carrier being slidably attached to the right side vertical support by a first fixed channel on the base; wherein the first fixed channel is horizontal to the right side edge of the base; said carrier being slidably attached to the left side vertical support by a second fixed channel on the base; wherein the second fixed channel is horizontal to the left side edge of the base; and a rotational power source connected to a first end of the carrier by at least one gear that rotates and causes the first end of the carrier to move up and down in relation to the base while simultaneously causing the carrier to move forward and backward in relation the first fixed channel on the base and the second fixed channel on the base.
2. The spray can shaker of claim 1 wherein: the carrier further includes a first projection located on a right side of the carrier that engages the first fixed channel; and a second projection located on a left side of the carrier that engages the second fixed channel.
3. The spray can shaker of claim 1 wherein: said power source is pivotally connected to the first end of the carrier via the at least one gear and a fly wheel having a counterweight.
4. The spray can shaker of claim 1 wherein: said power source is a crank handle.
5. The spray can shaker of claim 1 further comprising: a stationary handle extending above the spray can shaker in a substantially centered position to provide a contact point for a user to hold the spray can shaker at a fixed position on a work surface.
6. The spray can shaker of claim 1 further comprising: a non-slip surface located on the bottom surface of the base to keep the base stationary during use.
7. The spray can shaker of claim 1 wherein: said carrier further comprises an open tubular-shaped body having a front end, a back end, and at least one middle clip located between the front end of the body and the back end of the body.
8. The spray can shaker of claim 7 wherein: said at least one middle clip further comprises an indented step that acts as a stop to engage a perimeter edge of a spray can.
9. The spray can shaker of claim 1 wherein: said carrier further comprises an open tubular-shaped body having a front end, a back end, an inner surface, and an outer surface; and at least one C-shaped clip located within the tubular-shaped body that is used to secure a spray can into the carrier.
10. The spray can shaker of claim 9 wherein: said at least one C-shaped clip further comprises an

indented step that acts as a stop to engage a perimeter edge of a spray can.
11. The spray can shaker of claim 9 wherein: said carrier further comprises an open tubular-shaped body having a front end, a back end, and a top clip located on the front end of the body and a bottom clip located on the back end of the body.
12. The spray can shaker of claim 11 wherein: said carrier further comprises at least one middle clip located between the front end of the body and the back end of the body.
13. The spray can shaker of claim 12 wherein: said at least one middle clip further comprises an indented step that acts as a stop to engage a perimeter edge of a spray can.
14. The spray can shaker of claim 11 wherein: said top clip further comprises an indented step that acts as a stop to engage a top perimeter edge of a spray can; and said bottom clip further comprises an indented step that acts as a stop to engage a bottom perimeter edge of a spray can.
15. The spray can shaker of claim 14 wherein: said at least one C-shaped clip further comprises an indented step that acts as a stop to engage a perimeter edge of a spray can.